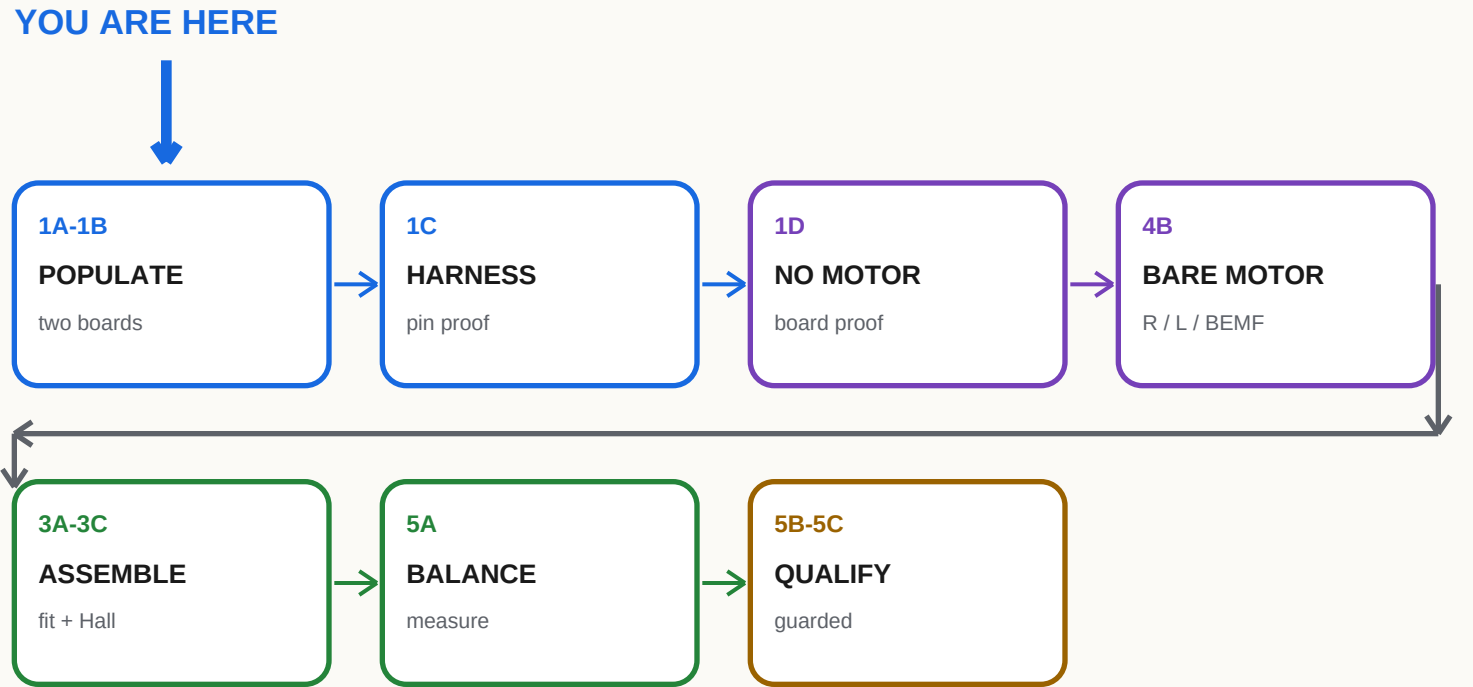




- WHAT COUNTS AS DONE
- ☐ A result is measured and recorded, not just observed.
 - ☐ Connector polarity and pin order are proved end-to-end.
 - ☐ Balance and hand clearance precede powered full-rotor work.
 - ☐ A failed gate stops the next powered stage.
 - ☐ Configuration must say verified before representative-rotor qualification.
 - ☐ 216 RPM proof uses a released guarded fixture and procedure.

BOUNDARIES KEPT OUT OF THIS ACTIVE BINDER

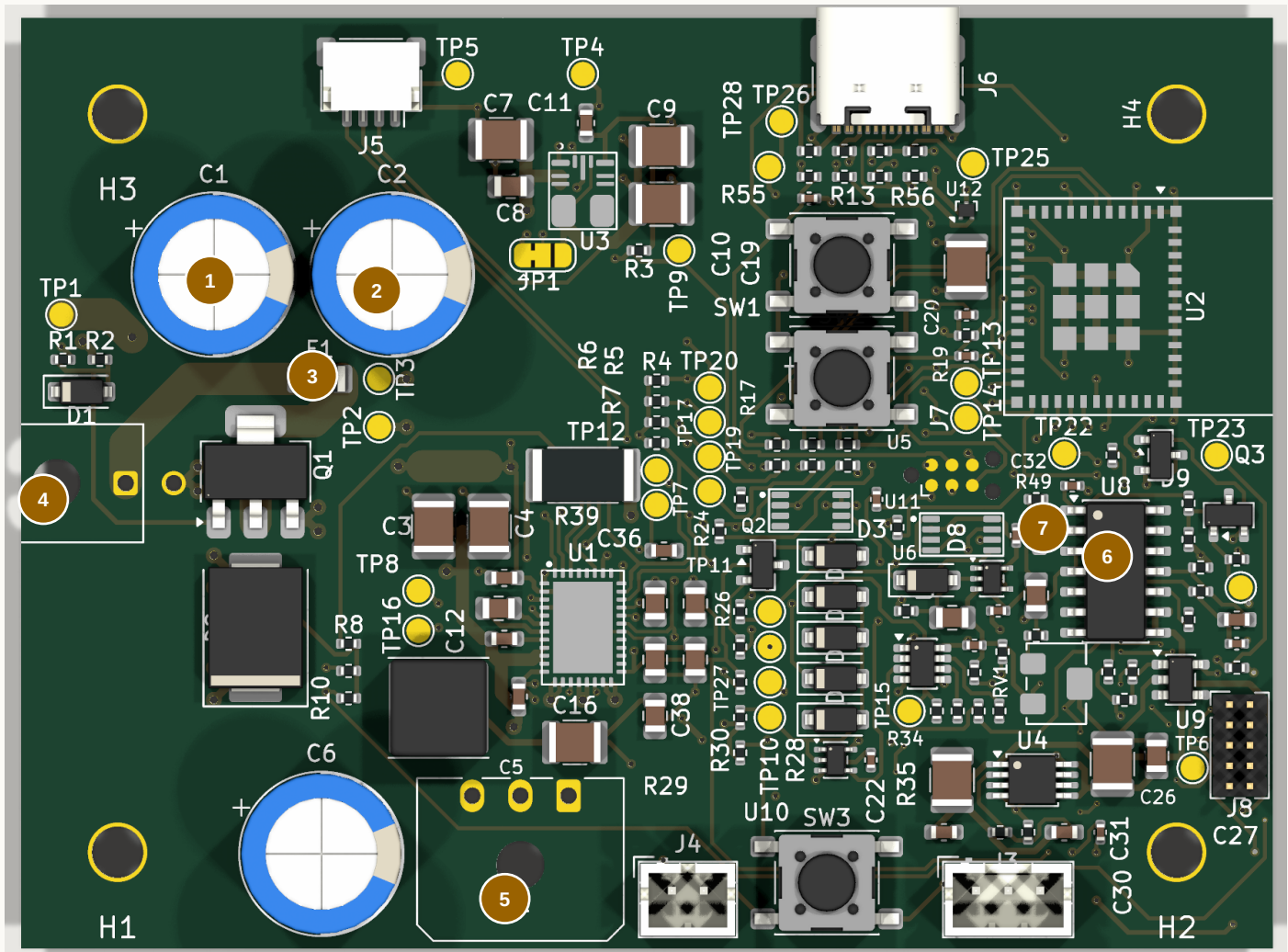
Owner-managed and deferred scope is intentionally omitted from this binder. It stays omitted unless explicitly reopened.

STOP / HANDOFF Start at 1A. Carry only the current sheet, board, and required tools to the bench.

1A PCB-01 Population Map

FIELD GUIDE

TOP / COMPONENT SIDE - KiCad target appearance



1 C1
5 J2

2 C2
6 U8

3 F1
7 C34

4 J1

SOLDER ORDER: U8 -> C34 -> J1/J2 -> C1/C2 -> INSPECT -> F1 LAST

1 + 2 C1 / C2

EEU-FR1H471, 470 uF 50 V. Positive lead to board +; negative stripe to pin 2 / PGND. Seat square without crushing bung.

3 F1 - INSTALL LAST

Low-profile tinned copper link across the 1206 pads. Near-zero ohms after soldering. External wall-side 3 A fuse remains mandatory.

4 + 5 CONNECTORS

J1 43045-0200; J2 43650-0300. Larger chisel tip. Tack one electrical pin, square and fully seat housing, then finish. Locating pegs are not soldered.

6 U8 / 7 C34

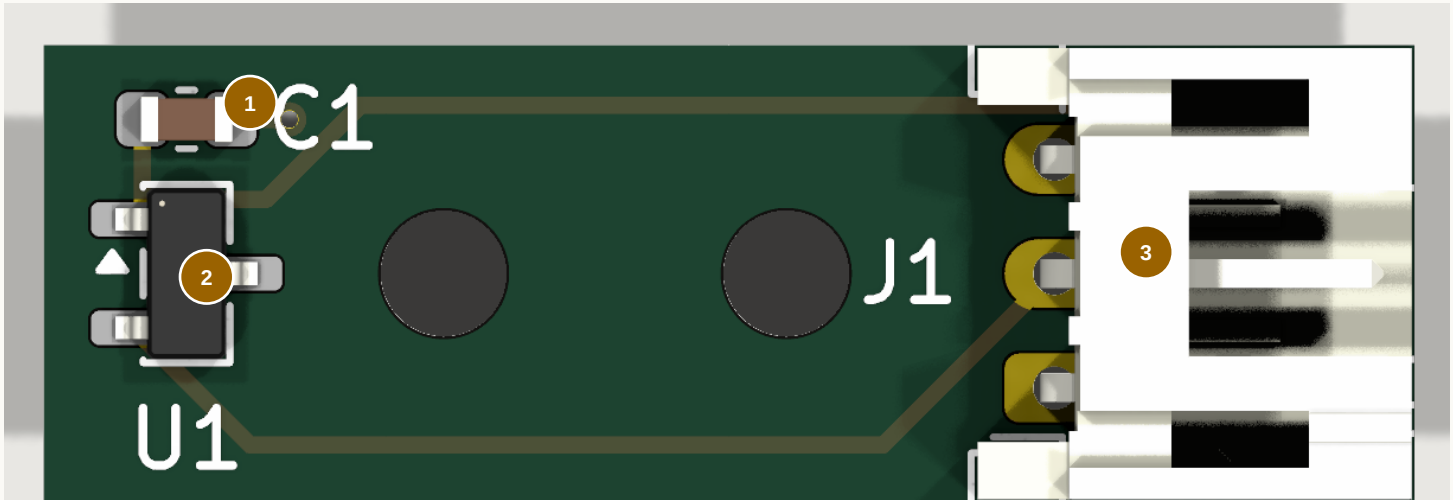
U8 LM2907M/NOPB: notch/dimple to pin-1 marker, tack pin 1 then diagonal pin 8, flux and drag-solder. C34 C1206C104K3GACTU: 100 nF COG 25 V +/-10%, spanning the smaller 0603 site, not shorting it.

PCB-01: USE THE IRON, NOT BROAD HOT AIR. Then clean and dry; inspect under the microscope; verify no hard short from RAW24, VM24, 3V3, or 12V_TACH to ground; photograph the board.

1B PCB-02 Population + Soldering

FIELD GUIDE

TOP / MAGNET-FACING COMPONENT SIDE - KiCad target appearance



J1 / CABLE EXITS RIGHT →

1 C1

C0603C104K5RACTU, 100 nF 50 V X7R, 0603. Pin 1 / 3V3 is left; pin 2 / AGND is right.

2 U1

DRV5033FAQDBZR, SOT-23. Match package index to board triangle. Pin 1 upper-left = 3V3; pin 2 lower-left = HALL_TACH; pin 3 right = AGND.

3 J1

S3B-PH-K-S side-entry JST-PH. Top to bottom in this view: 3 AGND, 2 HALL_TACH, 1 3V3. Iron-solder last.

NO HOT PLATE: CONTROLLED HOT-AIR METHOD

Tiny paste on U1 + C1

Place under microscope

Warm board gradually

Low-airflow reflow

Cool + inspect

Iron-solder J1

PASTE

Only enough to wet each pad, never fill the gaps. Follow the paste maker's reflow profile; the station setpoint is not joint temperature.

HOT AIR

Low airflow. Warm the whole small board gradually, then approach from above. Let surface tension center U1. Shield plastic and let the board cool before rework.

IRON ALTERNATIVE

Flux, tack U1 pin 3, correct alignment, then solder pins 1 and 2. For J1, tack one pin, square the body, finish, then revisit the tack.

FINAL INSPECTION

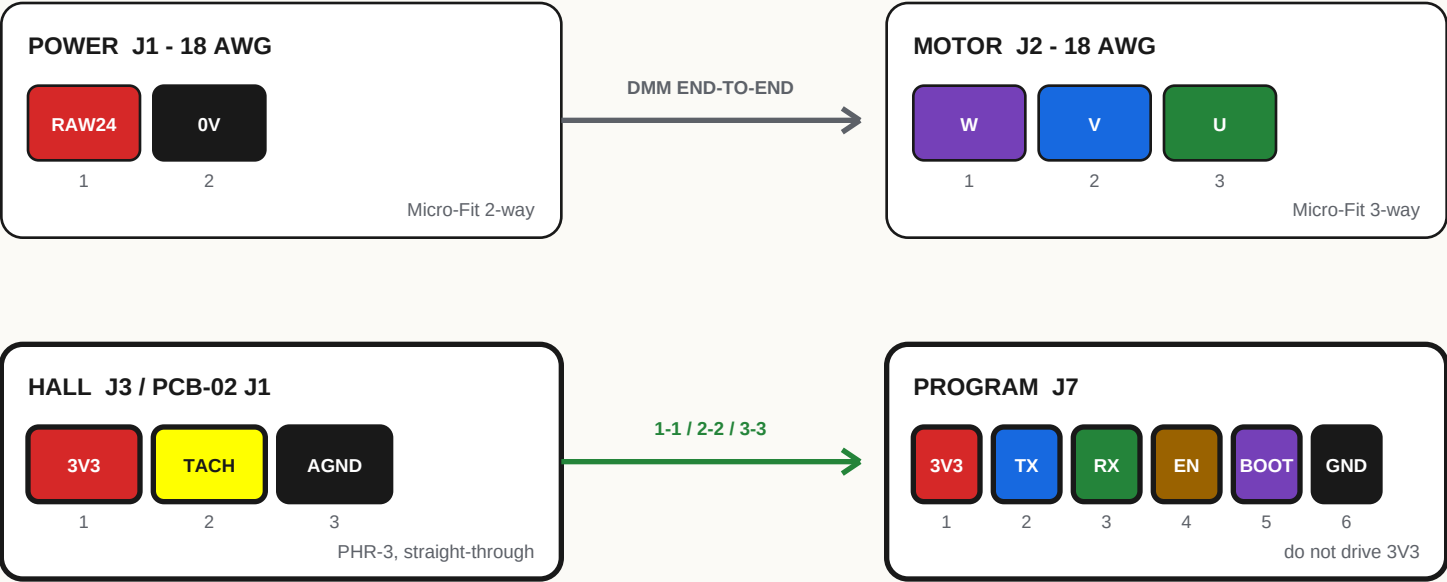
- ☐ All parts on magnet-facing side; sensor marked face will face magnet. Do not use magnet polarity as the orientation cue.
- ☐ Flux cleaned using its specified method; board completely dry.
- ☐ No short: 3V3 to HALL_TACH, 3V3 to AGND, or HALL_TACH to AGND.
- ☐ Microscope: wet fillets, no lifted lead, whisker, ball, or disturbed part; photograph board.

STOP / HANDOFF

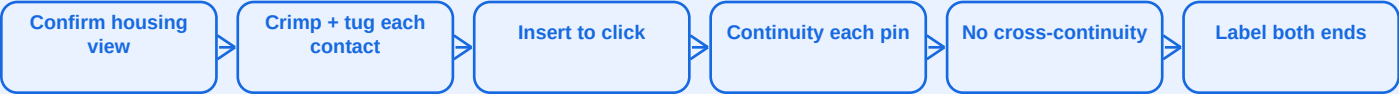
PCB-02 passes magnified inspection and unpowered three-net short checks. Do not power yet.

1C Harness Build + Pin Proof

LOOK INTO THE MATING FACE; USE MOLDED PIN-1 MARKS, NOT WIRE COLOR MEMORY



EACH HARNESS: BUILD -> PROVE -> LABEL



- ☐ Power: +24 V and 0 V preserved end-to-end.
- ☐ Motor: label positions W / V / U even if flying leads are not yet identified.
- ☐ Hall: exactly 1-to-1, 2-to-2, 3-to-3.
- ☐ Record length, gauge, wire colors, contact orientation, and result before sleeving.

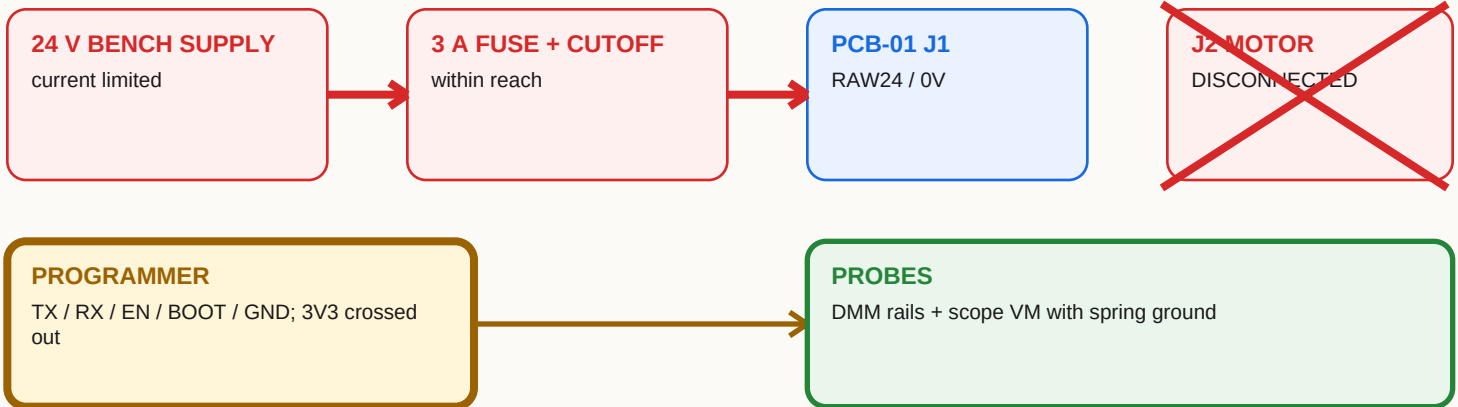
J7 PIN 1 IS BOARD 3V3, NOT A POWER INPUT. Never let the programmer drive it while 24 V is connected.

STOP / HANDOFF Every harness is labeled, pull-tested, and signed off for pin continuity and no cross-continuity.

1D PCB-01 First Power, No Motor

FIELD GUIDE

WIRE THIS FIRST; PLACE PROBES BEFORE POWER



NUMBERED TEST FLOW

1 RAILS

Brief power: 24 V, 3.3 V, AVDD, DVDD, DRVOFF; record current and heat.

2 HARDWARE CUTS

Force PGOOD, WDO, manual clear, OVERSPEED_N low one at a time -> DRVOFF rises without ESP help.

3 WATCHDOG

Static WDI: timeout 1.44-1.76 s, WDO pulse about 200 ms. 2 Hz falling edges prevent timeout.

4 STARTUP

Cold + slow-ramp power-up 20 times: U6 /PRE releases only after TACH_PGOOD_N.

5 RECOVERY

Remove/restore 24 V: disabled ≥ 10 s; requires new command plus explicit arm.

6 TACH

Bridge disabled. Inject 0-3.3 V, 1-3% duty, settle ≥ 30 s: reset ~ 3.00 Hz, trip ~ 3.33 Hz.

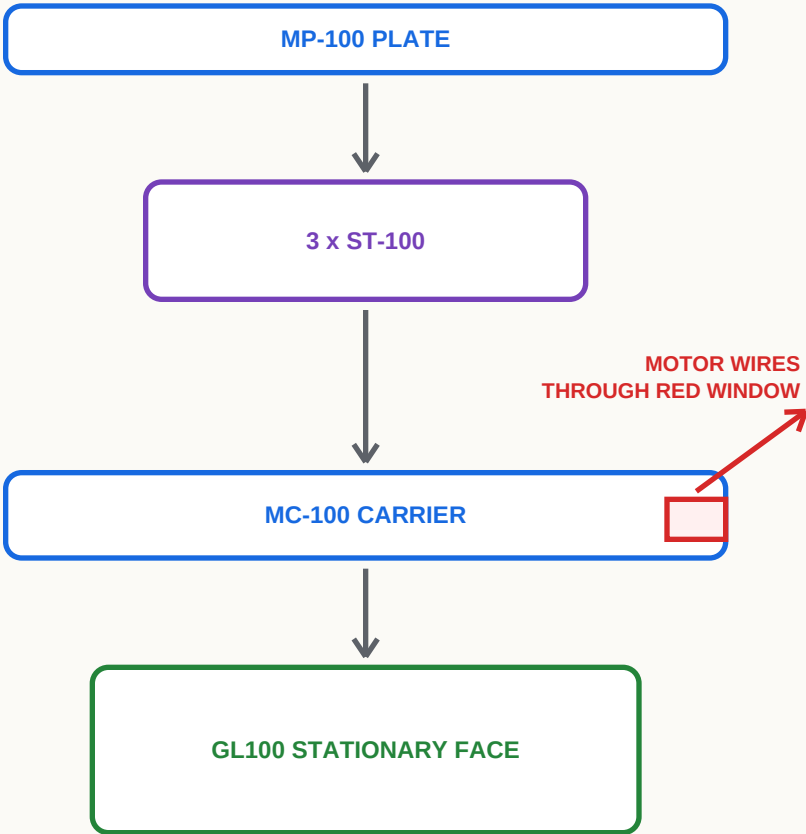
PASS LIMITS: VM ≤ 35 V target; 40 V rejects. Persistent tach lock clears only on a genuine low-voltage power cycle. Any automatic re-arm blocks release.

STOP / HANDOFF

All listed board/safety tests have recorded passes. Do not connect the motor after any failure.

3A Stationary Stack Dry-Fit

EXPLODED SIDE VIEW - NON-POWERED



MP-100 -> ST-100
3 x M6 x 16 flat-head from plate ceiling face; heads flush or below.

ST-100 -> MC-100
3 x M6 x 20 A4-80 + Nord-Lock pairs. Hand-start, square seating.

MC-100 -> GL100
4 x M4 x 12 A4-80 into the 60 mm stationary pattern. The 50 mm face rotates.

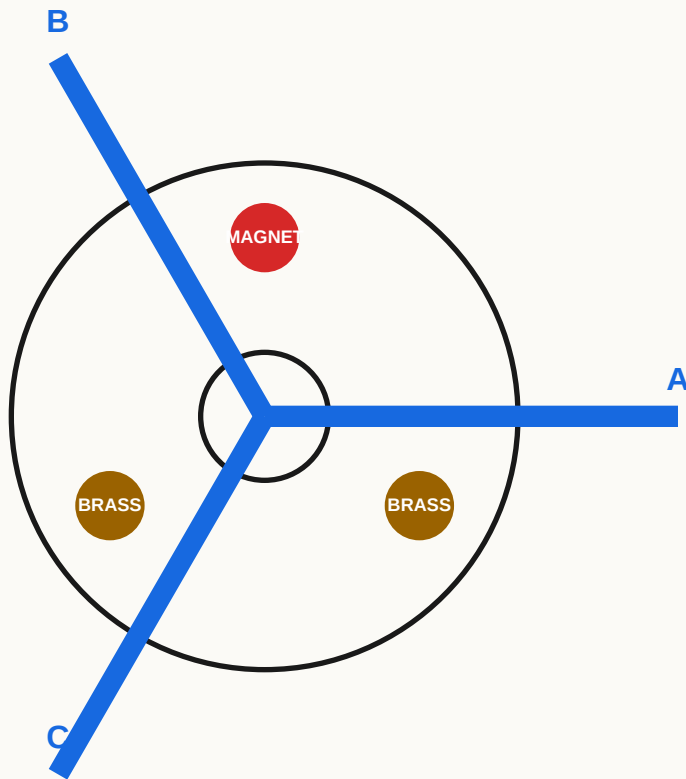
THREAD DEPTH
Target about 5.5 mm engagement; 6.0 mm motor-thread maximum. A bottoming screw is felt at hand torque: stop and correct.

RELEASE CHECK

- | | |
|---|---|
| ☐ All fasteners hand-start; no rocking or forced alignment. | ☐ Motor wire is unpinched and clear of sharp edges. |
| ☐ Owner-selected torque applied consistently; witness marks present. | ☐ Threadlocker only where specified; no screw bottoms. |
| ☐ Stationary 60 mm and rotating 50 mm faces confirmed. | ☐ Stack remains non-powered until later gates pass. |

STOP / HANDOFF Dry stack sits square without force, rocking, wire pinch, or bottoming screws.

3B Rotor + Tach Inserts



HUB TO MOTOR

4 x M4 x 10 A4 flat-head from underside into 50 mm rotating pattern; heads 0.1-0.2 mm subflush. Pilot enters bore by hand only.

EACH BLADE

BP-100 v3 locating pins + captured M5 nuts; 4 x M5 x 22 per blade. Torque consistently and witness-mark.

THREE INSERT STATIONS

Epoxy retains one magnet + two brass slugs. Match complete insert-plus-epoxy station masses within 0.01 g after cure.

MASS + FINAL RECORD

MAGNET _____ g BRASS B _____ g BRASS C _____ g MAX SPREAD _____ g

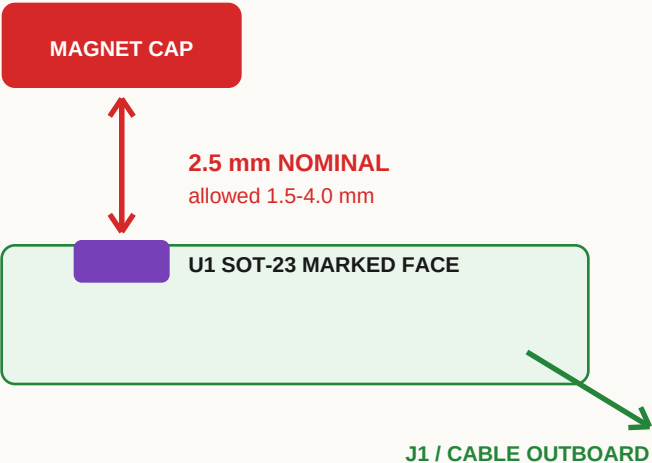
- ☐ Pilot hand-fit; all four hub heads subflush.
- ☐ Epoxy fully cured; inserts cannot move.
- ☐ Critical fasteners torqued and witness-marked.
- ☐ All three blade stations have four correct bolts and captured nuts.
- ☐ Hand-rotate several revolutions: no rub, click, or changing clearance.
- ☐ Photograph rotor before balance measurements.

Do not use cad/BP-100.step for current fit; it is stale v2 geometry.

STOP / HANDOFF

Complete rotor turns freely with correct retention; continue to 5A before powered motion.

SECTION THROUGH MAGNET AND SENSOR



COMMON CENTERLINE

Magnet and sensor element at radius 76.0 +/-0.5 mm.

PCB-02 MOUNTING

M2 holes on 6 mm pitch. At H1 use bare pan head or washer <=4.5 mm OD. Components face magnet.

BR-100

Strain-relieve Hall cable. Validate bracket around physical motor, board, magnet, and real gap.

PCB-01 BRACKET ENVELOPE

110 x 80 x 25 mm SERVICE ENVELOPE

Four 6-8 mm M3 standoffs. Keep isolated mounting holes clear of circuit ground. Independent PCB retention and cable clamps. Preserve connector and cable-bend keepouts.

HAND-ROTATION RELEASE

- ☐ One clean pulse/revolution with magnet.
- ☐ All cables and connector bodies clear rotor.
- ☐ No false pulse with magnet removed.
- ☐ Gap remains in range through full hand rotation.

STOP / HANDOFF

Hall pulse is clean; retained PCB and clamped cables clear all moving parts.

4A Flash + One Persistent CLI Session

FIELD GUIDE

BUILD / FLASH / TALK

```
cargo build --manifest-path firmware/Cargo.toml
```

```
esptool flash --port /dev/cu.usbmodem2101 --non-interactive  
firmware/app/target/riscv32imac-unknown-none-elf/debug/stillair
```

```
firmware/target/debug/stillair --port /dev/cu.usbmodem2101 state
```

BARE DEV-BOARD INPUT JUMPERS

GPIO22 -> 3V3

PGOOD good

GPIO21 -> 3V3

nFAULT idle high

GPIO14 -> GND

ALARM idle low

SafeBoot

Starting

15 s, no FG

NoRotation fault

EXPECTED ON A BARE BOARD; IT DOES NOT GENERATE FG EDGES

SESSION RULES

ONE READER

Stop raw-log capture before esptool or CLI opens the port.

USE SCRIPT

Multi-step simulator work must stay in one session; separate invocations reset it.

RELEASE GATE

config check must report config=verified. ok=true plus unverified is not a pass.

wait <state> --for <seconds> after asynchronous commands | stream <hz> --for <seconds> for CSV | config capture only after complete measured configuration

Never use raw reg write on a spinning motor except under an approved stopped/instrumented procedure. The simulator validates the harness, not the motor.

STOP / HANDOFF

Firmware flashes; one persistent CLI session communicates and preserves scripted state.

4B Bare GL100 Measurements

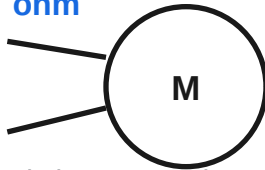
HOLD: IMAGE UNVERIFIED

NO BLADES x

Guarded motor; manual cutoff ready; board/safety tests already passed

1 PHASE RESISTANCE

ohm



method: _____ ohm

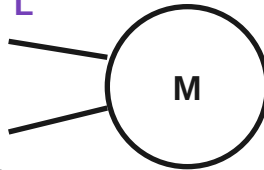
U-V: _____

V-W: _____

W-U: _____

2 PHASE INDUCTANCE

L



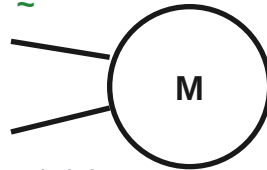
frequency: _____ L

U-V: _____

V-W: _____

W-U: _____

3 MANUAL-SPIN BEMF



RPM/polarity: _____ Vpp

U-V: _____

V-W: _____

W-U: _____

Repeat across phase pairs; record connection and polarity convention. Confirm 20 pole pairs.

GREEN LANE - DO NOW

☐ Measure R and L independently with method recorded.

☐ Treat register seeds 0xB1 / 0xAE / 0xCA as provisional.

☐ Manually spin only; scope line-to-line BEMF and convention.

☐ Keep loose hub hardware and blades off the motor.

AMBER LANE - HOLD

☐ No unloaded MPET; representative final rotor required.

☐ No config apply release until EEPROM completion/readback is proved.

☐ Golden image needs measured D-generation fields and review.

☐ Release requires config check: verified, not merely ok=true.

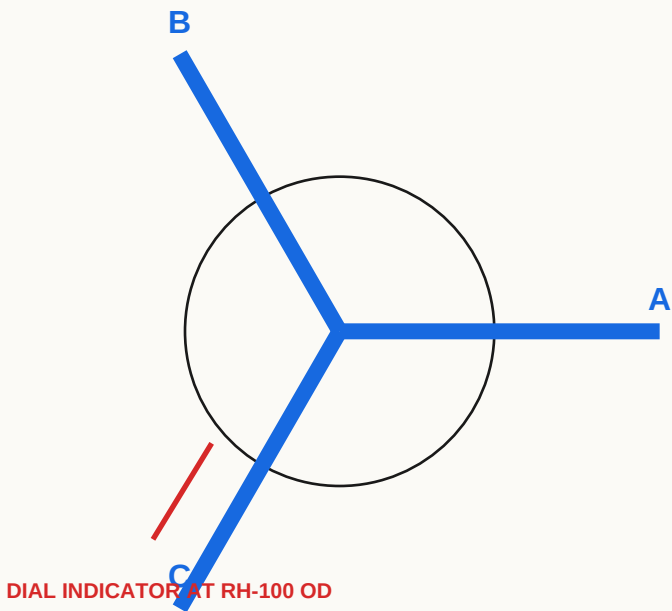
No live raw register writes. For any later powered work, VM <=35 V target; 40 V rejects.

STOP / HANDOFF

R / L / BEMF records are complete. Loaded MPET and the golden image remain HOLD.

5A Rotor Balance + Runout Record

FIELD GUIDE



HUB RUNOUT
<=0.10 mm TIR

BLADE TIP HEIGHT
all three in one indexed setup; spread <=0.5 mm

FIRST MOMENT
each blade; spread <=0.5%

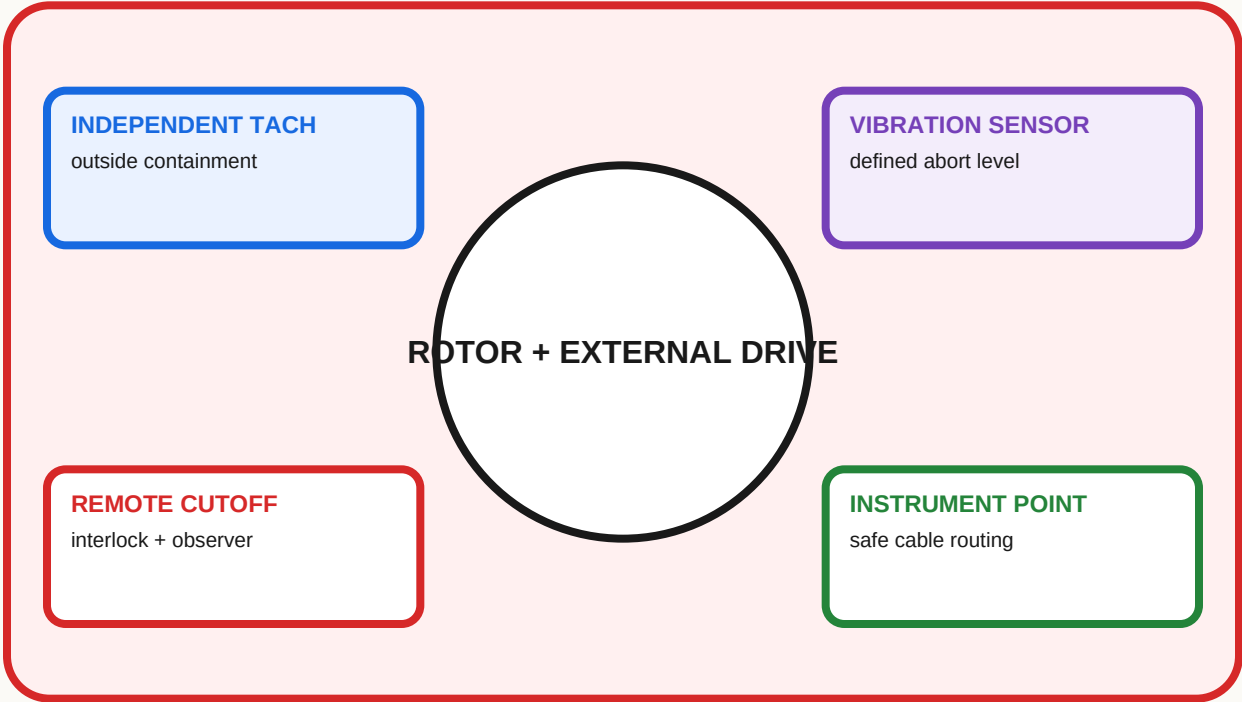
A / B / C MEASUREMENT TABLE

STATION	MASS (g)	FIRST MOMENT	TIP (mm)	CORRECTION
A				
B				
C				

METHOD CHECK

- ☐ 0.01 mm indicator + 0.01 g calibrated scale.
- ☐ Define vibration rejection before powered motion.
- ☐ Correct only by documented slug method; remeasure.
- ☐ Hand-clearance pass and witness-mark photo saved.

STOP / HANDOFF MEC-05 passes numerically; do not advance a rotor that only looks balanced.



OPERATOR A SAFE POSITION

OPERATOR B SAFE POSITION



270 RPM = CALCULATION ONLY x NEVER DYNAMICALLY TEST

RELEASE BEFORE RUNNING

- ☐ Rated containment/fixture drawing released.
- ☐ Exact bypass state and restoration checks written.
- ☐ Pre-run balance, runout, clearance, witness marks recorded.
- ☐ Two-person roles, callouts, safe positions, duration defined.
- ☐ Independent RPM and measurable vibration/contact abort set.
- ☐ After bypass, reverify 180 RPM MCF and 200 RPM analog limits.

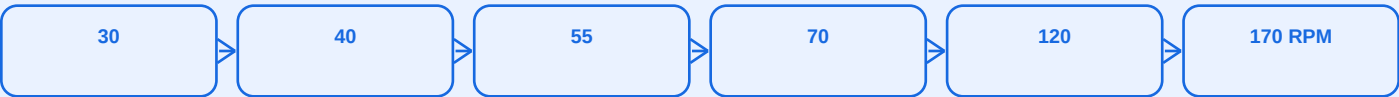
STOP / HANDOFF This sheet does not authorize the run. Release fixture and procedure first.

START MATRIX - PASSES / TARGET

DIRECTION	23.3 V (5)	24.0 V (20)	24.7 V (5)
CW			
CCW			

REJECT: retry, reverse kick, stall, hunting, or objectionable tonal sequence

SPEED LADDER



MCF active-control ceiling <=180 RPM; config must be verified before this stage

8-HOUR THERMAL RUN @ 170 RPM

	0h	1h	2h	3h	4h	5h	6h	7h	8h
AMBIENT C									
MOTOR C									
PCB C									
RMS A									
INPUT W									
ANOMALY									

CURRENT

<0.8 A normal | 1.0 A investigate | 1.5 A limiter not continuous

TEMPERATURE

motor <70 C | PCB <85 C

POWER

input <50 W; no dropout/overtemp

STOP / HANDOFF Starts, speed ladder, essential shutdowns, cooldown inspection, and thermal run all pass.